ApexPlanet Cybersecurity Internship

Task 1: Foundation & Environment Setup

Full Report

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Task	Task 1: Foundation & Environment Setup (Days 1-12)
LinkedIn	https://www.linkedin.com/in/rupashi-sethi-4505a936a/
GitHub repository	https://github.com/rupa-code/Task-1--Apexplanet-cybersecurity-intern
Video (LinkedIn post)	https://lnkd.in/p/dA2vqEM9

Introduction

This report is my submission for Task 1 of the ApexPlanet Cybersecurity & Ethical Hacking Internship. The task asks for strong basics in cybersecurity, networking and cryptography, and a working hacking lab. I studied each topic, built a private lab with Kali Linux and Metasploitable2 in VirtualBox, tested it, and recorded a short walkthrough video. All lab screenshots in this report come from my own machines.

Task Deliverables

Deliverable	Where to find it
Lab Setup Report (screenshots of Kali, Metasploitable and a Wireshark test capture)	Section 2 of this report (Figures 1 to 6)
GitHub repository with the Task 1 report	github.com/rupa-code/Task-1--Apexplanet-cybersecurity-intern: this report and screenshots
Video walkthrough of the lab setup	LinkedIn post: https://lnkd.in/p/dA2vqEM9

1. Cybersecurity Basics

1.1 The CIA Triad

The CIA Triad is the base of information security. Every control we use protects at least one of these three goals.

Principle	What it means	Common protections	Real-life example
Confidentiality	Only authorised people can see the data.	Encryption, strong passwords, access control.	Your bank balance is visible only to you and the bank.
Integrity	Data stays accurate and cannot be changed without permission.	Hashing, digital signatures, version control.	A student's marks cannot be edited by an outsider.
Availability	Systems and data are ready when needed.	Backups, backup servers, disaster recovery plans.	An online shop stays open during a sale.

1.2 Common Threat Types

Threat	What it is	How to reduce the risk
Phishing	Fake emails, messages or websites that trick people into giving passwords or card details.	Check the sender, do not click unknown links, turn on MFA.
Malware	Harmful software such as viruses, worms, trojans and spyware.	Use antivirus, keep software updated, download from trusted sources.
DDoS	Thousands of devices flood a server so real users cannot reach it.	Traffic filtering and DDoS protection services.
SQL Injection	Malicious SQL typed into a form to read or change a database.	Parameterised queries and input validation.
Brute Force	Trying many passwords or keys until one works.	Long passwords, account lockout, MFA.
Ransomware	Locks or encrypts files and asks for money.	Offline backups, updates, careful with attachments.

1.3 Attack Vectors

- **Social engineering:** attackers trick people instead of breaking systems. Examples are pretexting (a made-up story), baiting (a tempting offer) and tailgating (following someone into a building).
- **Wireless attacks:** fake Wi-Fi hotspots (evil twin), sniffing packets on open Wi-Fi, and cracking weak Wi-Fi passwords.
- **Insider threats:** a person inside the company who misuses access. They may be malicious, careless, or have had their account taken over.

2. Lab Environment Setup

A safe lab lets me practise security tools without touching any real network. My lab runs entirely inside VirtualBox on my laptop.

Part	Role in the lab
VirtualBox	Runs the virtual machines.
MahaKali (Kali Linux)	Attacker machine that comes with security tools such as Wireshark, Nmap and Burp Suite.
Target metasploitable2	Intentionally vulnerable machine used as the practice target.

Part	Role in the lab
Host-Only network	Private network shared only between my VMs and my laptop. It has no route to the internet or other devices.

2.1 Steps I Followed

1. Installed VirtualBox and created the Kali Linux VM named MahaKali.
2. Imported the Metasploitable2 disk and created the target VM.
3. Set Metasploitable2 to a **Host-Only Adapter**. MahaKali has a Host-Only adapter and a NAT adapter.
4. Logged in to both machines and checked their IP addresses with **ifconfig**.
5. Tested the link by sending **ping** packets from Kali to Metasploitable2.
6. Captured the ping traffic with **Wireshark**.
7. Saved a snapshot called "Clean Lab Baseline" so I can restore the lab at any time.

2.2 Screenshots

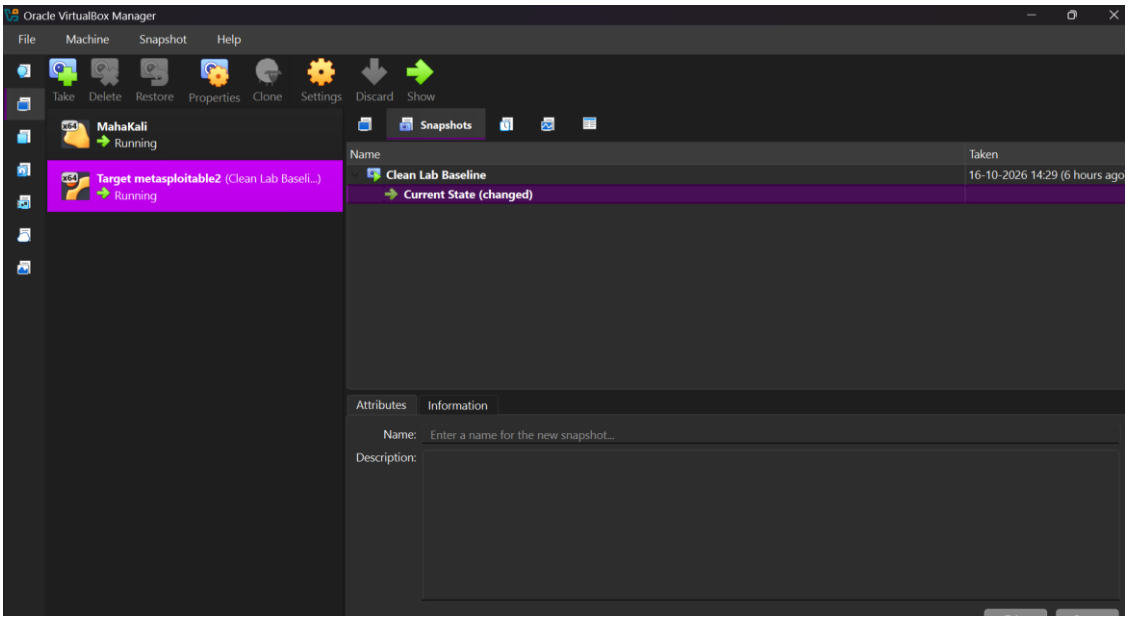


Figure 1: VirtualBox Manager with both VMs running and the snapshot "Clean Lab Baseline".

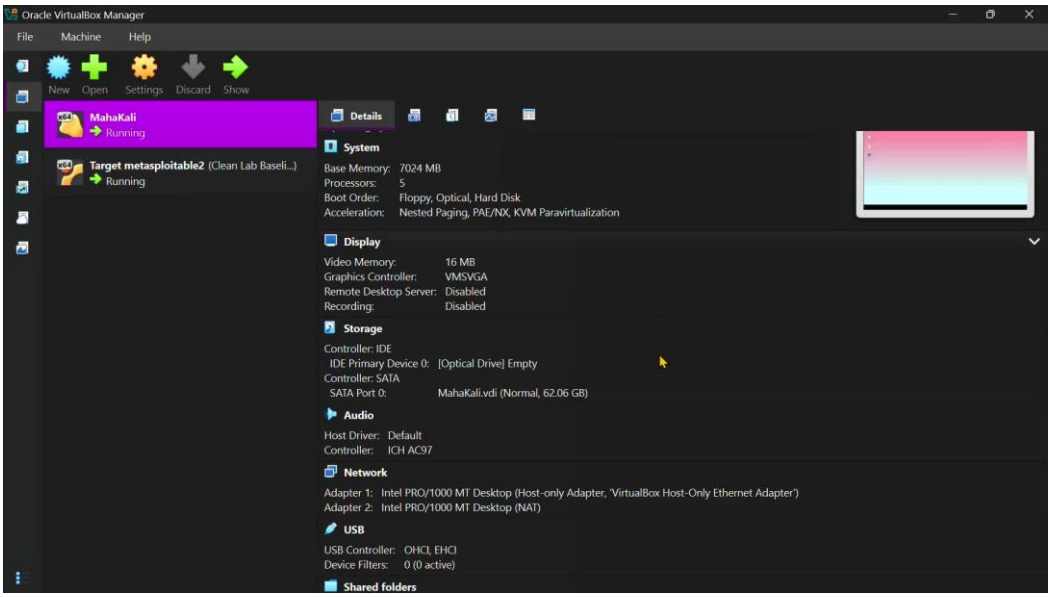


Figure 2: MahaKali network settings: Adapter 1 = Host-only, Adapter 2 = NAT.

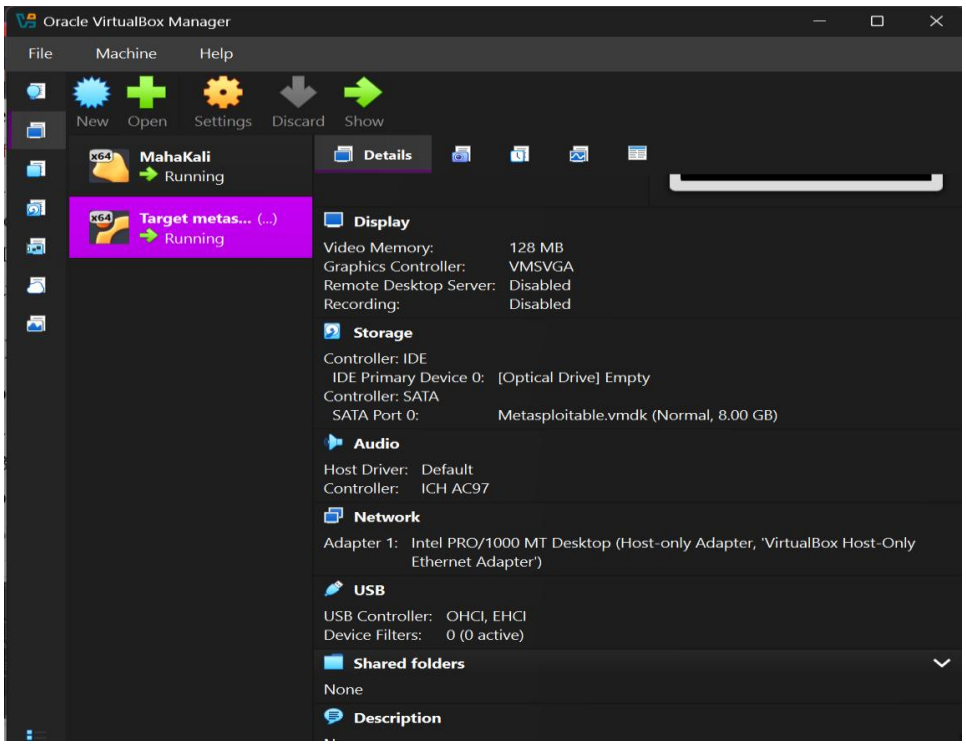


Figure 3: Metasploitable2 network settings: Adapter 1 = Host-only Adapter.

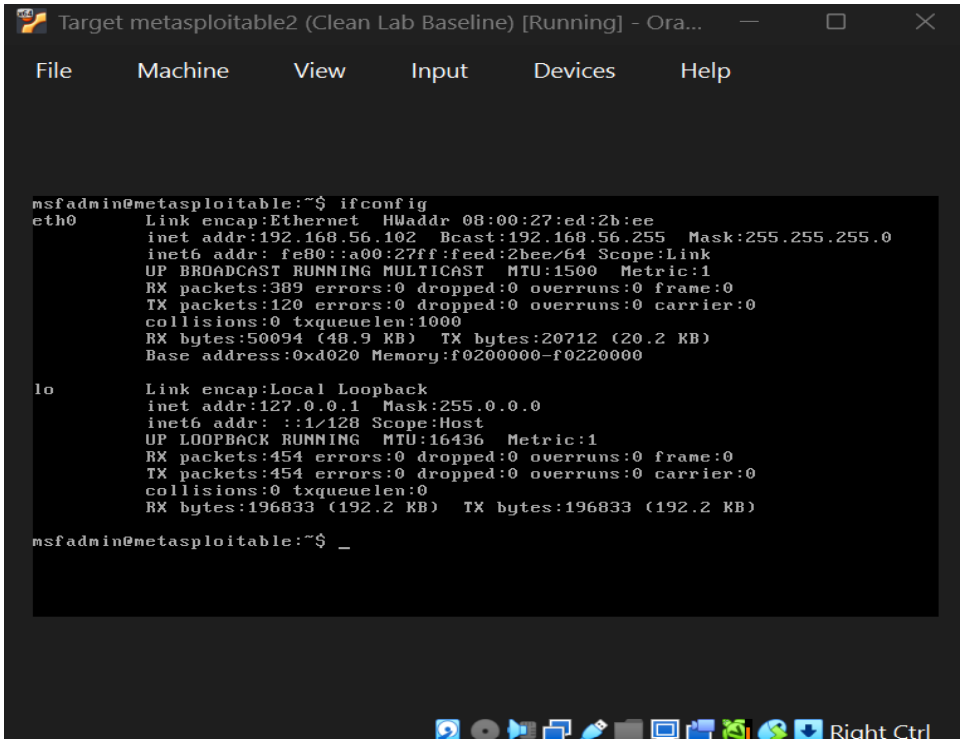


Figure 4: ifconfig on Metasploitable2: IP 192.168.56.102, MAC 08:00:27:ed:2b:ee.

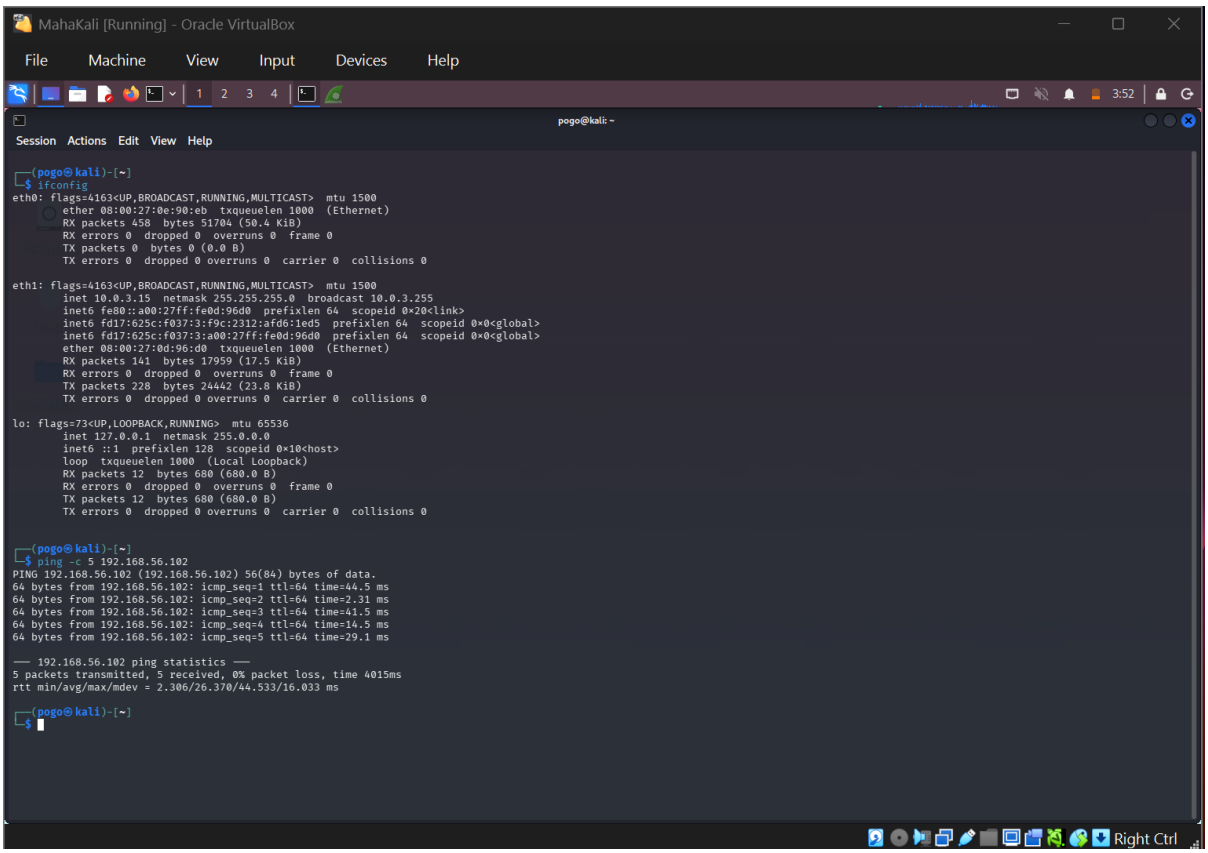


Figure 5: ifconfig on Kali (eth1 = 10.0.3.15) and ping -c 5 to 192.168.56.102: 5 sent, 5 received, 0% packet loss.

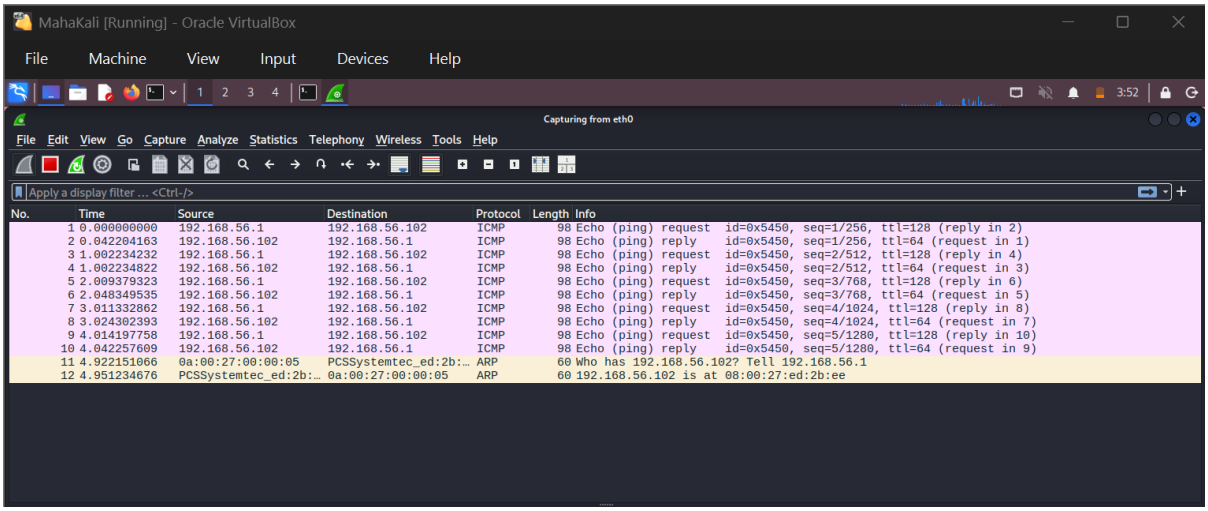


Figure 6: Wireshark test capture on eth0 showing ICMP echo requests and replies plus ARP packets.

2.3 Results and Observations

Check	Result
Metasploitable2 IP (eth0)	192.168.56.102
Kali IP (eth1)	10.0.3.15
Ping, Kali to Metasploitable2	5 sent, 5 received, 0% packet loss. Time: min 2.3 ms, average 26.4 ms, max 44.5 ms.
Wireshark capture	ICMP echo request and reply pairs between 192.168.56.1 and 192.168.56.102, followed by ARP packets.

- **Reply TTL values:** replies from Metasploitable2 have TTL 64 (typical for Linux). Requests from 192.168.56.1, my laptop on the Host-Only network, have TTL 128 (typical for Windows).
- **ARP in action:** the capture shows "Who has 192.168.56.102? Tell 192.168.56.1", and the answer "192.168.56.102 is at 08:00:27:ed:2b:ee". This is how a device finds the MAC address that belongs to an IP address on its local network.
- **Safety:** Metasploitable2 is full of known weaknesses, so it stays on the Host-Only network only.

3. Linux Fundamentals

Command	What it does	Example
pwd	Prints the folder you are in.	pwd
ls	Lists files. -l shows details, -a shows hidden files.	ls -la
cd	Changes folder. cd .. goes one level up.	cd Documents
chmod	Changes permissions: read (4), write (2), execute (1).	chmod 755 script.sh
chown	Changes the owner and group of a file.	sudo chown kali:kali notes.txt
apt	Installs, updates and removes software from online repositories and solves dependencies.	sudo apt update; sudo apt install nmap
dpkg	Installs or inspects a local .deb package file.	sudo dpkg -i package.deb
ifconfig	Shows IP address, netmask and MAC address of each network card.	ifconfig
ping	Checks whether a machine replies and how fast.	ping -c 4 192.168.56.102
netstat	Shows open ports and active connections.	netstat -tuln
traceroute	Shows each router a packet passes on its way to a destination.	traceroute google.com

Reading permissions: in chmod 755, the first digit is the owner (7 = 4+2+1, read, write, execute), the second is the group (5 = read and execute) and the third is everyone else (5 = read and execute).

4. Networking Basics

4.1 OSI Model

Layer	Job	Examples
7 Application	Gives network services to the user.	HTTP, DNS, FTP, SMTP
6 Presentation	Translates, encrypts and compresses data.	SSL/TLS, JPEG, ASCII
5 Session	Opens, manages and closes sessions.	NetBIOS, RPC
4 Transport	End-to-end delivery, ports, error recovery.	TCP, UDP
3 Network	Addressing and routing between networks.	IP, ICMP, routers
2 Data Link	Delivery between devices on the same network.	MAC address, switch
1 Physical	Sends raw bits as signals.	Cables, Wi-Fi radio

4.2 TCP/IP Suite

TCP/IP has four layers: Application, Transport, Internet and Network Access. At the Transport layer, TCP and UDP behave differently:

	TCP	UDP
Connection	Sets up a connection first.	No connection setup.
Reliability	Checks, re-sends and keeps order.	No guarantee of delivery or order.
Best for	Web pages, email, file transfer.	Video calls, streaming, games.

4.3 DNS and HTTP/HTTPS

How DNS finds a website:

1. The browser asks the DNS resolver (usually run by the ISP) for the IP of the domain.
2. If the resolver does not know, it asks a root server, which points to the TLD server (.com, .org).
3. The TLD server points to the authoritative server for that domain.
4. The authoritative server gives the IP address, and the resolver returns it to the browser.

	HTTP	HTTPS
Data	Sent as plain text.	Encrypted with TLS.
Port	80	443
Identity	Server is not verified.	Server proves its identity with a certificate.

4.4 IP Addressing, Subnetting and NAT

- **IP address:** IPv4 uses 32 bits (for example 192.168.56.102). IPv6 uses 128 bits. Private ranges are 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16. My lab uses 192.168.56.x.
- **Subnetting:** splits a large network into smaller ones for better security and less traffic. Example: 192.168.1.0/24 has 256 addresses (254 usable hosts). Two /25 subnets give 126 usable hosts each.
- **NAT:** lets many private devices share one public IP address. PAT does this by giving each connection a different port number.

5. Cryptography Basics

5.1 Symmetric vs Asymmetric Encryption

	Symmetric	Asymmetric
Keys	One shared secret key.	A public key and a private key.
Speed	Fast, good for large data.	Slower.
Main problem	Sharing the key safely.	Slow for big files.
Examples	AES, DES, ChaCha20	RSA, ECC, Diffie-Hellman
Used for	Disk, file and VPN encryption.	Key exchange, digital signatures.

5.2 Hashing (MD5 and SHA-256)

A hash turns any data into a fixed-size fingerprint. It works one way only, and changing even one character gives a totally different hash. The table shows the hashes of the text "Hello World".

Algorithm	Size	Hash of "Hello World"	Status
MD5	128 bit	b10a8db164e0754105b7a99be72e3fe5	Weak, collisions are possible. Only for simple checksums.

Algorithm	Size	Hash of "Hello World"	Status
SHA-256	256 bit	a591a6d40bf420404a011733cfb7b190d62c65bf0bcda32b57b277d9ad9f146e	Strong. Used for integrity, signatures and TLS.

5.3 Digital Certificates and SSL/TLS

A digital certificate, issued by a trusted Certificate Authority, proves that a website is genuine. In a TLS connection the browser checks the certificate, uses asymmetric encryption to agree on a session key, and then uses fast symmetric encryption for the actual data. This is what the padlock in the browser means.

5.4 Encrypting and Decrypting with OpenSSL

OpenSSL can encrypt a message with a password using AES-256. The commands below show the method (reference):

```
echo "Hello Apex" > secret.txt
openssl enc -aes-256-cbc -salt -pbkdf2 -in secret.txt -out secret.enc
openssl enc -d -aes-256-cbc -pbkdf2 -in secret.enc -out decrypted.txt
```

The first command creates a text file, the second encrypts it (it asks for a password), and the third decrypts it with the same password.

6. Tool Familiarization

Tool	Purpose	Example command or use	Status in this task
Wireshark	Captures and shows network packets.	Capture on eth0 while pinging.	Used in the lab (Figure 6).
Nmap	Finds live hosts and open ports.	nmap 192.168.56.102	Studied.
Burp Suite	Web proxy to inspect and test web requests.	Browser proxy set to 127.0.0.1:8080.	Studied.
Netcat	Reads and writes data over network connections.	nc -lvp 4444 (listen), nc host 4444 (connect)	Studied.

Wireshark was the tool I used hands-on. Its capture on eth0 (Figure 6) shows how a ping works at packet level: an ICMP echo request goes out, an echo reply comes back, and ARP first finds the MAC address of the target.

Conclusion

In Task 1 I learned the core ideas of cybersecurity, networking, Linux and cryptography, and built a working practice lab. Kali Linux is my attacker machine, Metasploitable2 is my target, and both sit on an isolated Host-Only network with a clean snapshot saved. I proved the lab works with ping and a Wireshark capture, documented it with screenshots, and recorded a walkthrough video. This gives me a safe base for the tasks that follow.